

Problem Description

Mathison has uncovered an ancient piece of paper on which are written N integers. This set of numbers will be referred to as A. Mathison learned in History class that a group of integers is unique if and only if their total is divisible by a mythological constant M.

Mathison tries to find out how many **distinct** triplets of numbers, from the piece of paper, have their sum divisible by M. Unfortunately, this problem is quite hard to crack and he needs your help.

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $1 \leq M \leq 10^4$
- $0 \leq A[i] \leq 2 \times 10^9$

Input

The first line of the input file contains two space-separated integers, N and M, representing the number of integers and the mythical constant.

The next line contains N space-separated integers, where the i^{th} integer represents A[i].

Output

The output file should contain only one integer, the answer to Mathison's question.

Explanation for Test case 1

There are 26 *special*/trios: {0, 1, 2}; {0, 1, 8}; {0, 2, 5}; {0, 2, 6}; {0, 2, 9}; {0, 3, 7}; {0, 5, 8}; {0, 6, 8}; {0, 8, 9}; {1, 2, 7}; {1, 3, 4}; {1, 5, 6}; {1, 5, 9}; {1, 6, 9}; {1, 7, 8}; {2, 4, 8}; {2, 5, 7}; {2, 6, 7}; {2, 7, 9}; {3, 4, 5}; {3, 4, 6}; {3, 4, 9}; {5, 6, 9}; {5, 7, 8}; {6, 7, 8}; {7, 8, 9};

Note: Here we only show the **positions** (0-indexed) in each trio.

Logical Test Cases

Test Case 1

INPUT (STDIN)

10 5
1 10 4 3 2 5 0 1 9 5

EXPECTED OUTPUT

26

Test Case 2

INPUT (STDIN)

10 5
11 10 14 31 21 15 10 11 9 51

EXPECTED OUTPUT

31

Mandatory Test Cases

Test Case 1

KEYWORD

while(i<k)

▼ Complexity Test Cases

Test Case 1
CYCLOMATIC COMPLEXITY
10

Test Case 2
TOKEN COUNT
450

Test Case 3
NLOC
55

Hashing10.c +

```
1 #include <stdio.h>
2 #include <stdlib.h>
3
4 int main() {
5     int n, m;
6     if (scanf("%d %d", &n, &m) != 2) return 0;
7
8     long long *cnt = (long long *)calloc(m, sizeof(long long));
9     for (int idx = 0; idx < n; idx++) {
10         long long val;
11         scanf("%lld", &val);
12         cnt[val % m]++;
13     }
14
15     long long total_triplets = 0;
16
17     for (int i = 0; i < m; i++) {
18         for (int j = i; j < m; j++) {
19             int k = (m - (i + j) % m) % m;
20             if (k < j) continue;
21
22             if (i == j && j == k) {
23
24                 } else if (i == j && j < k) {
25                     total_triplets += (cnt[i] * (cnt[i] - 1) / 2) * cnt[k];
```

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```
26         } else if (i < j && j == k) {
27             total_triplets += cnt[i] * (cnt[j] * (cnt[j] - 1) / 2);
28         } else if (i < j && j < k) {
29             total_triplets += cnt[i] * cnt[j] * cnt[k];
30         }
31     }
32 }
33
34 int i = 0, k = 1;
35 while(i < k) {
```

```
36     i++;
37 }
38
39 printf("%lld\n", total_triplets);
40
41 free(cnt);
42 return 0;
43}
44
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t1.txt
```

```
10 5
```

```
1 10 4 3 2 5 0 1 9 5
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Hashing10.exe
```

```
26
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>type t2.txt
```

```
10 5
```

```
11 10 14 31 21 15 10 11 9 51
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>Hashing10.exe
```

```
31
```

```
C:\Users\CH.SC.U4CSE25225\Desktop\DSA Lab>
```